

REVIEW 1

Literature Survey · Research Gap · References

Smart Inventory & Warehouse Management System

Team Members

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Under the Guidance of: Course Faculty Guide

Agenda

- Recap
- Literature Survey
- Research Gap
- Proposed Direction
- References

Recap — Review 0

- Problem: manual, spreadsheet-based inventory tracking is slow and error-prone.
- Motivation: a centralized database improves accuracy and saves staff time.
- Objectives: design a relational schema and automate stock updates and alerts.

Literature Survey

- Comparative review of common inventory-tracking approaches, not a peer-reviewed paper survey — informs the design of the proposed DBMS solution.

Approach Reviewed	Description	Limitation
Spreadsheet-based tracking	Excel/Google Sheets used to log stock manually.	No relational integrity; slow, error-prone at scale.
Barcode-based inventory systems	Barcode scanning updates stock counts on sale.	Needs scanner hardware; still lacks relational reporting.
Basic POS software	Point-of-sale tools track sales, minimal stock logic.	Little support for multi-warehouse or supplier data.
Cloud ERP/SaaS inventory tools	Commercial platforms (e.g., Zoho, Odoo modules).	Costly, generic, and not customizable for course needs.
RFID-based tracking	Radio tags automate item-level identification.	High hardware cost; overkill for small/mid warehouses.

Research Gap

- Existing spreadsheet and POS-based methods lack enforced relationships between products, suppliers, and warehouses.
- Low-stock detection is manual in most small-scale tools — no automatic reorder flagging.
- Commercial ERP/SaaS options are costly, generic, and difficult to customize or extend for a specific business.
- Very few lightweight, self-hosted solutions combine a normalized relational schema with a simple web interface.
- **Gap**
- No low-cost, database-driven, multi-warehouse solution with automated alerts

Proposed Direction

- **Relational core:** MySQL schema with enforced primary/foreign-key relationships across all entities.
- **Automation:** stock levels update automatically on every transaction; low-stock items are flagged.
- **Accessible interface:** Flask + Bootstrap web app so staff never write raw SQL.
- **Efficient retrieval:** joins, aggregate functions, views, and stored procedures for fast reporting.

References

#	Reference
1	Elmasri, R., & Navathe, S. B. — Fundamentals of Database Systems.
2	MySQL 8.0 Reference Manual — Oracle Corporation, official documentation.
3	Flask Documentation — Pallets Projects, official documentation.
4	Silberschatz, A., Korth, H., & Sudarshan, S. — Database System Concepts.
5	Bootstrap Documentation — responsive UI framework reference.
6	Comparative study of inventory management approaches (course literature review).